How To Reduce Size And Weight Of PULSE LOAD POWER SUPPLY

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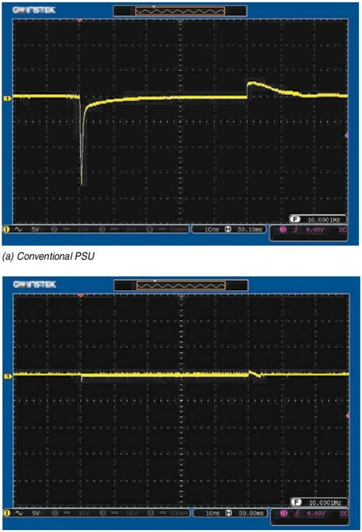
There are a number of electronics devices that require clear, pulsed power supply. For instance, power consumption of contemporary high-performance processor devices directly depends on the dynamic load of the processor and can vary over a very wide range within a short time.

Apart from this, transceiver equipment consumes the most power in transmitting mode, while in receiving mode there is virtually no load on the equipment. Such dynamics of load puts quite strict requirements in the output power quality of power supplies with step-change nature of output current. Limited voltage feedback speed of power supply units (PSUs) requires capacitive accumulators to be installed between them and the dynamic load. At the same time, you can see an unambiguous dependence between dynamic characteristics of a source and the value of a necessary capacitance accumulator.

It is evident that slow feedback requires a larger capacitance accumulator, which increases the weight and dimensions of the complete power supply system. The miniaturisation trend in electronics makes it necessary to increase the operating speed of power supplies to reduce the size of capacitance accumulator with the same quality of power supply to the load.

Given that, it is reasonable to use PSUs with very fast voltage feedback. This will reduce the number of capacitors required, and in some cases completely elimi-



(a) Conventional PSU

(b) PSU with fast voltage feedback

Fig. 1: Transient output voltage deviation of two different PSUs. Nominal output voltage 50 VDC. Step change of load 0W-340W-0W. Scale: Vertical 5V/div, horizontal - 10ms/div